CAPACITY PLANNING / PERFORMANCE ENGINEERING

DESIGN FOR LOAD BEFORE YOU NEED IT

Viral moments don't announce themselves. Load testing, horizontal scaling strategies, auto-scaling policies, and caching layers are investments that pay off exactly once — in the moment everything is on the line.

KEY BENEFITS

- ✓ Converts surprise traffic spikes from outages into growth events
- ✓ Reveals bottlenecks (databases, caches, third-party APIs) before users do
- ✓ Auto-scaling reduces infrastructure cost during normal load while handling peaks
- ✓ Builds team confidence to pursue growth without fear of the platform failing

REAL-WORLD EXAMPLE

The 2018 FIFA World Cup streaming service suffered massive outages because load testing had not kept pace with actual growth projections. Brand damage and subscriber losses lasted far beyond the incident.

PRACTICAL EXAMPLE

Every sprint, the team runs a 30-minute load test at 10× current peak. This week they discover their session store saturates at 8× — they switch to Redis Cluster before it ever becomes a user problem.

REFLECTION QUESTION

What is the maximum load your system has been tested against? How does that compare to a realistic peak scenario?